

Shaun Michael Verch

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Work	Site Reliability Engineer, Oso HQ • Core maintainer of the Rust authorization server and storage management • Migrated to AWS ECS on EC2 to decouple storage from compute • Ran replicated storage across 13 regions for edge reads • Served authorization queries at 2000 peak QPS, 40ms 99th percentile • Developed instance draining using Autoscaling Termination Policies	August 2023 - Present
	Freelance SRE Consultant • Consulting on architecture, reliability, cloud migrations, and security	June 2022 - July 2023
	SRE Consultant, Oso HQ • Supported initial launch of Oso Cloud, authorization as a service	February 2022 - June 2022
	Site Reliability Engineer, U.S. Digital Service • Led quiet vaccines.gov Covid-19 vaccine search tool launch • Built and open sourced a ServiceNow API testing framework	December 2020 - February 2022
	Site Reliability Engineer, PlanetScale • Automated deployment of Kubernetes, supporting infrastructure on AWS, GCP, Azure • Built Prometheus based monitoring stack and created incident response process • Built safe draining into the Kubernetes operator, verified by randomized state machine testing	January 2019 - November 2020
	Creator, Cloudless • Built cloud agnostic deployment prototype for AWS and GCP • Explored cloud provider data models for resource schema mapping	July 2018 - November 2020
	Site Reliability Engineer, U.S. Digital Service • Launched login.gov on AWS, Terraform, and Chef, now supporting 5 million users • Supported quiet launch of qpp.cms.gov across 6 remote DevOps teams	July 2016 - July 2018
	Site Reliability Engineer, URX • Managed 5+ engineers and 30+ services in zero downtime datacenter upgrade • Served 875M total queries, under 60ms average, 400ms at the 99th • Supported PySpark and 181TiB HDFS infrastructure used by ML engineers • Created lock free high performance persistent crawler queue	August 2014 - April 2016
	Database Systems Engineer, MongoDB • Core maintainer of the MongoDB server, wrote “willitlink” to fix dependency problems	August 2012 - June 2014
	Storage Engineering Intern, NetApp • Wrote automation and enhanced SCSI support for NetApp disk arrays	June-August 2011, 2012
Degrees	B.S. in Mathematics / Computer Science - Brown University, Providence, RI M.Sc. in Computer Science - Brown University, Providence, RI	
Books	Co-author, <i>MongoDB in Action, 2nd edition</i> - Manning Publications	
Teaching	<u>Conference Talks:</u> OSCON, MongoDB Days, Scale 11x, NoSQL Now <u>Online Class:</u> M101JS: MongoDB for Node.js developers <u>Teaching Assistant:</u> Operating Systems, Computing, Multivariable Calculus	
Software	<u>Languages:</u> Python, Rust, Golang, C, C++, Bash, Ruby, Java, Scala, Javascript, Intel x86 assembly, MIPS assembly, Perl, PHP <u>Operating Systems:</u> Linux, FreeBSD, MacOS, WSL <u>Datastores:</u> MySQL, Postgres, Vitess, MongoDB, Kafka, Elasticsearch, HBase, HDFS, Ceph, Zookeeper, Hadoop Mapreduce, Apache Spark <u>Operations:</u> AWS, GCP, Azure, Ansible, Tailscale, OpenVPN, Wireguard, Bind DNS, LDAP, sssd, Apache Mesos, Jenkins CI, Chef, Terraform, Pulumi, Kubernetes, Prometheus, Victoria Metrics, Honeycomb, Grafana, Datadog	